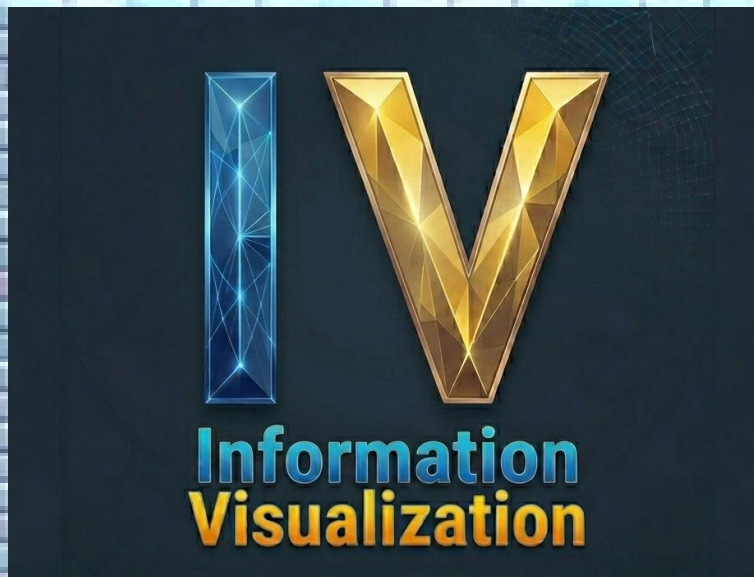


Vorlesung
Informationsvisualisierung



Kapitel 5 VISUALISIERUNG VON GRAPHEN



Lecture
Information Visualization

Chapter 5 VISUALIZATION OF GRAPHS

Next Tutorial

2

- Tuesday, 2.6.2026
 - ▣ Exercise 5
 - ▣ Exercise 6

Chapter
5.1

Applications

Harry C. Beck,
London Underground,
1933



<https://www.historyhit.com/in-pictures-the-evolution-of-the-london-tube-map/>

Tube map



MAYOR OF LONDON

Website
tfl.gov.uk

24 hour travel information
020 7222 1234

Transport for London



London Underground, today



Social Network Analysis

6

vizster

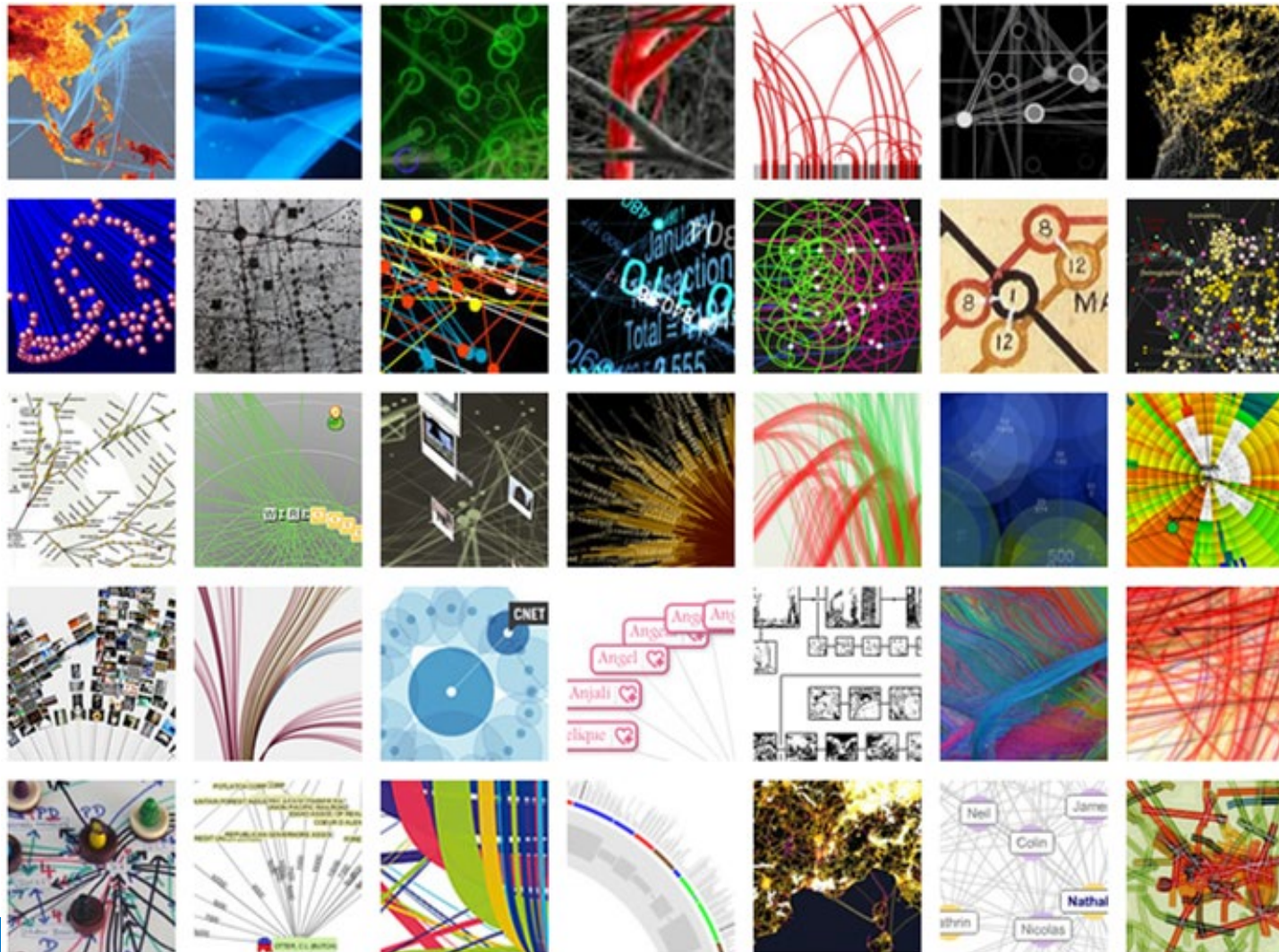


Jeffrey Heer,
Egocentric Social Network,
2004

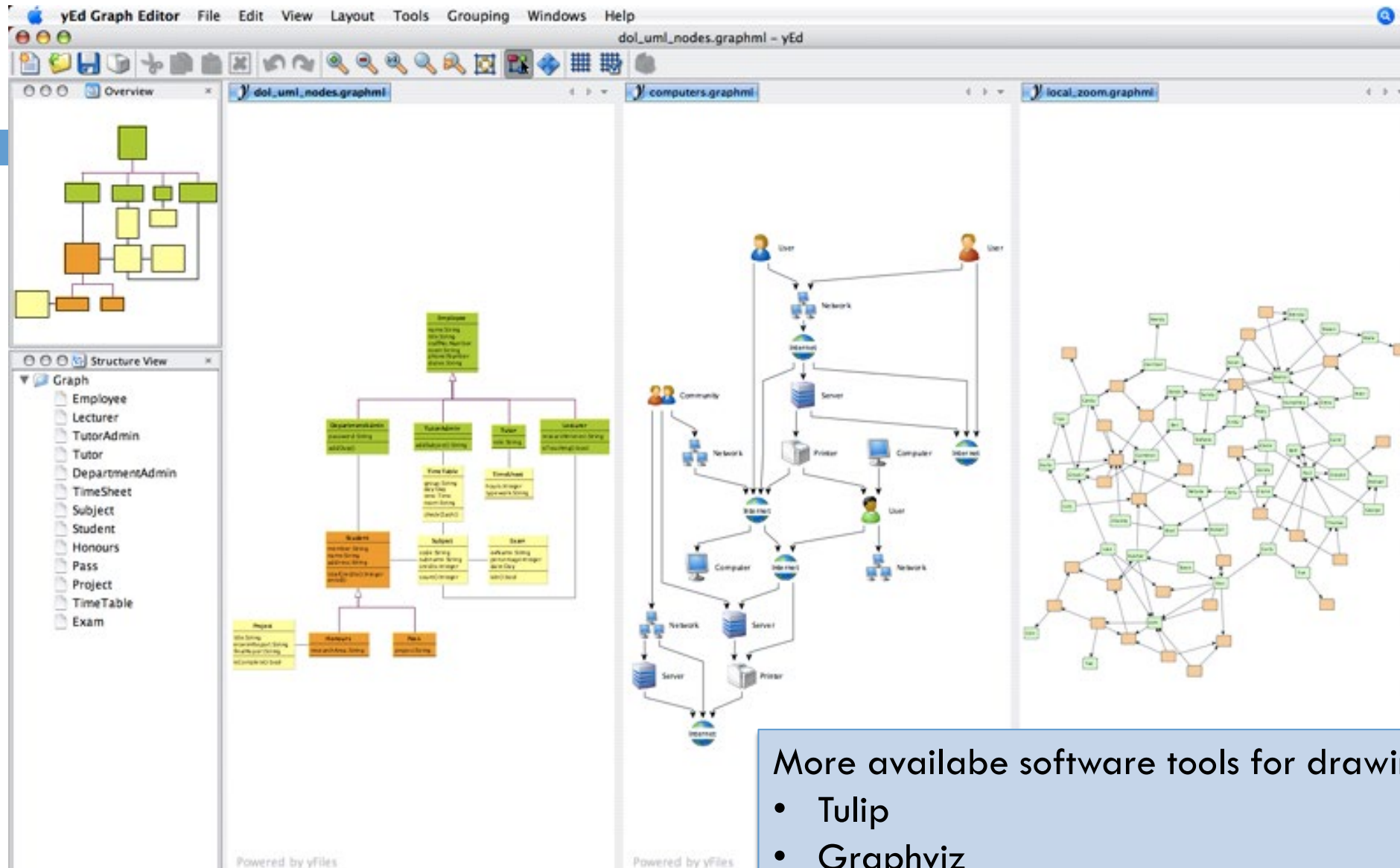


Visualization of Graphs

7



Graph Drawing Software: yEd



More available software tools for drawing graphs:

- Tulip
- Graphviz
- Gephi

Chapter
5.2

Definitions and Terminology

Definition: Graph

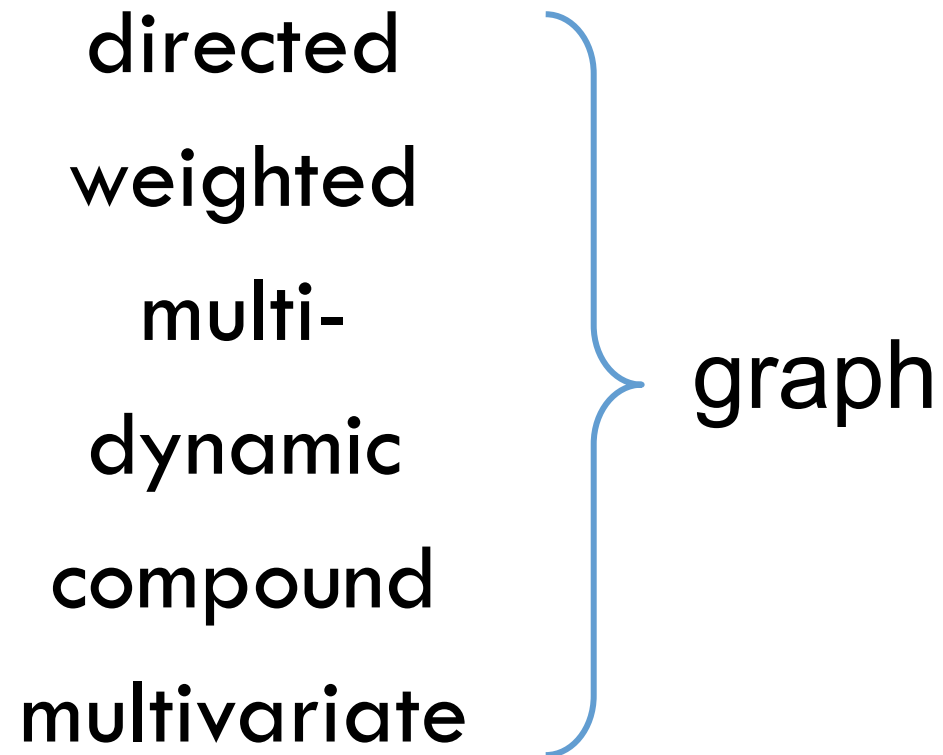
10

$$G = (V, E)$$
$$E \subset V \times V$$

A graph G consists of a set of vertices V which are connected by a set of edges E .

Types of Graphs

11



Properties of Graphs

12

tree

bipartite

acyclic

sparse/dense

complete

connected

Graph Drawing

13

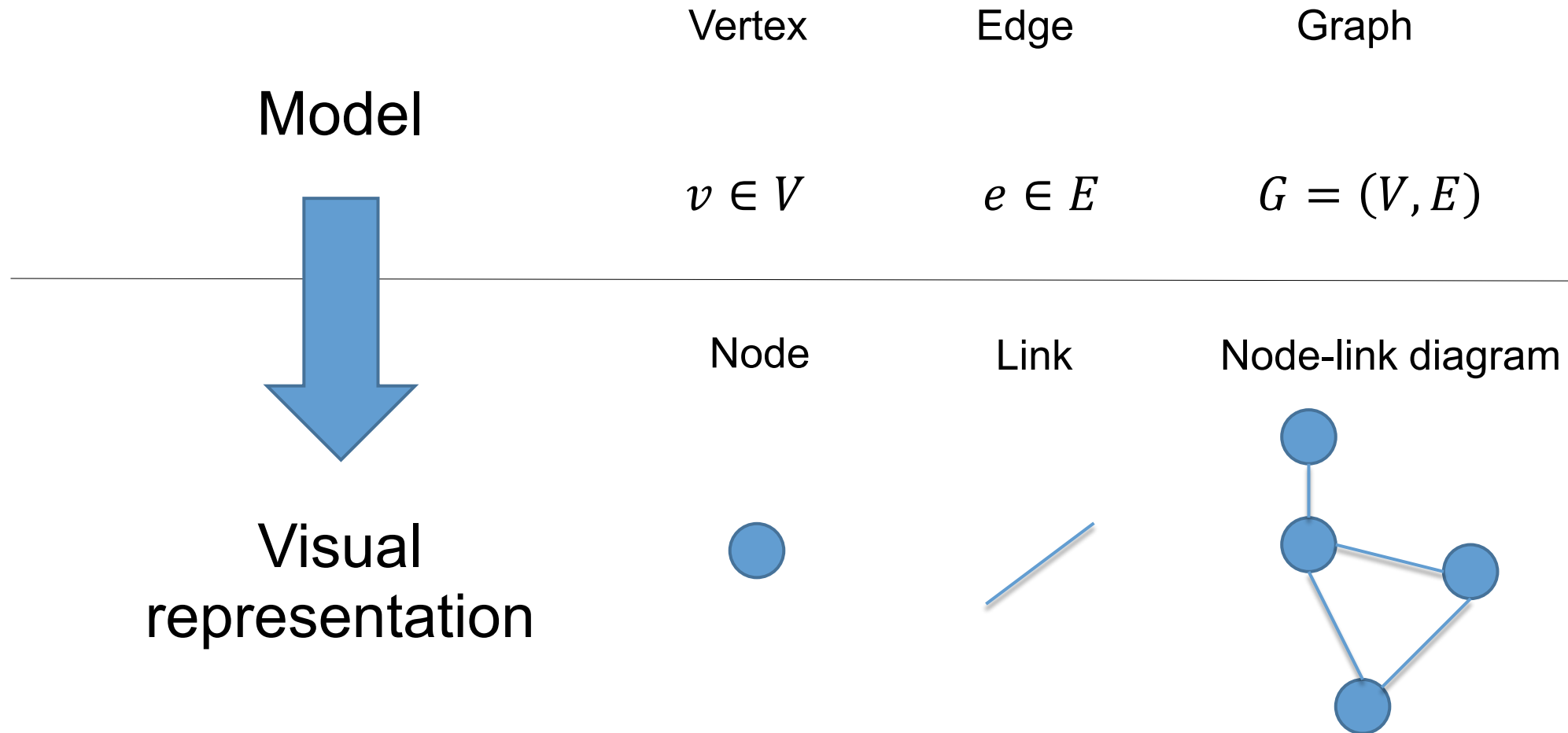
*The **art** of drawing a diagram of a graph to facilitate **understanding** of relations between objects.*

[Stephan Diehl, Software Visualization, Springer, 2007]

Node-Link Diagrams

Visual Mapping

15



Common visual encoding

16

edge weight $\Rightarrow$ thickness or color of link

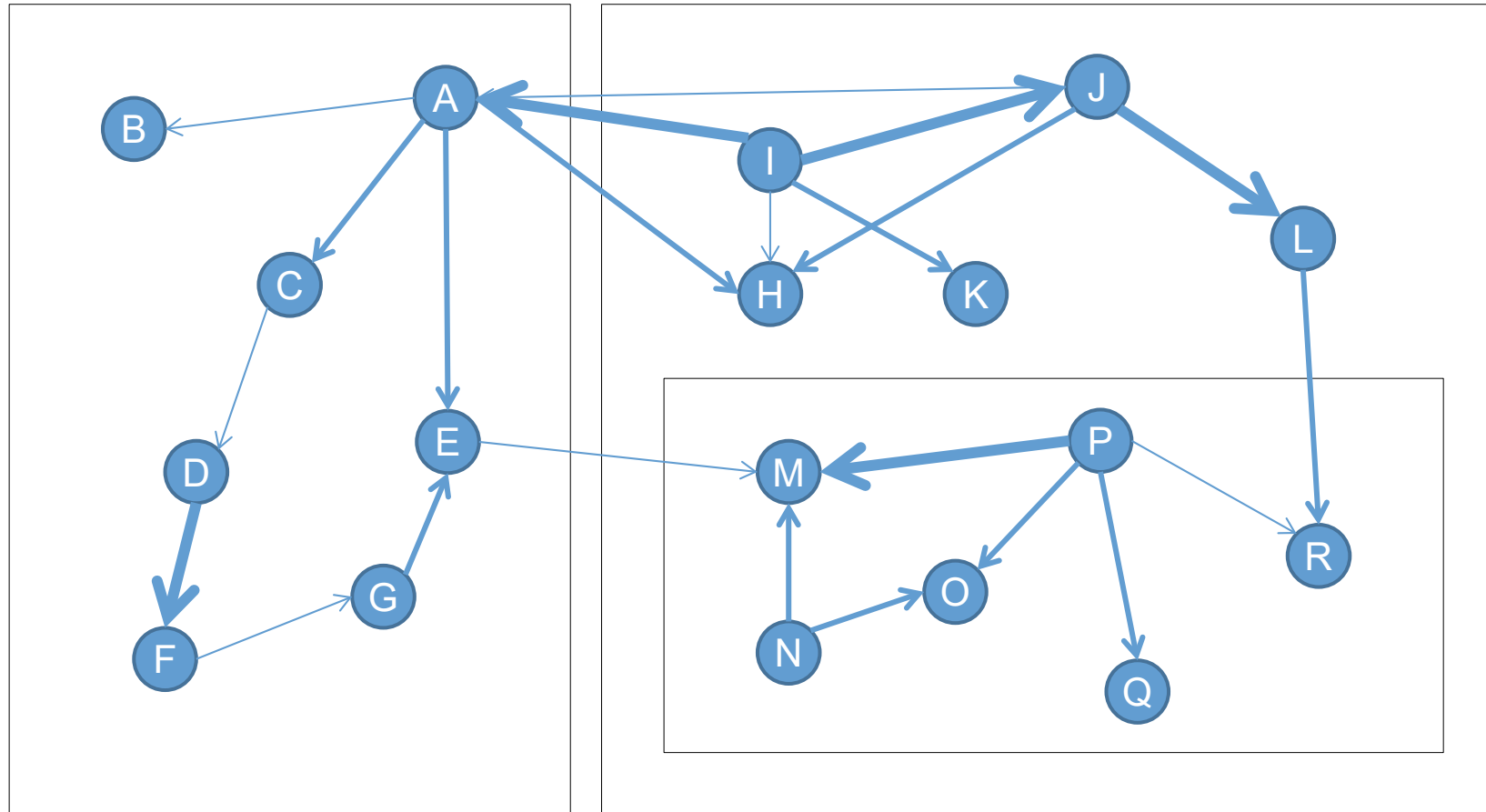
edge direction $\Rightarrow$ arrow head

hierarchy $\Rightarrow$ nested boxes

vertex names $\Rightarrow$ node labels

Example

17



weighted, directed compound graph

Chapter
5.3

Aesthetic Criteria



*Aesthetic Criteria for
drawing graphs?*

In empirical studies often: aesthetic = readable

Aesthetic Criteria

20

Minimize edge crossings

Minimize drawing area

Minimize edge length

Minimize number of bends (of edges)

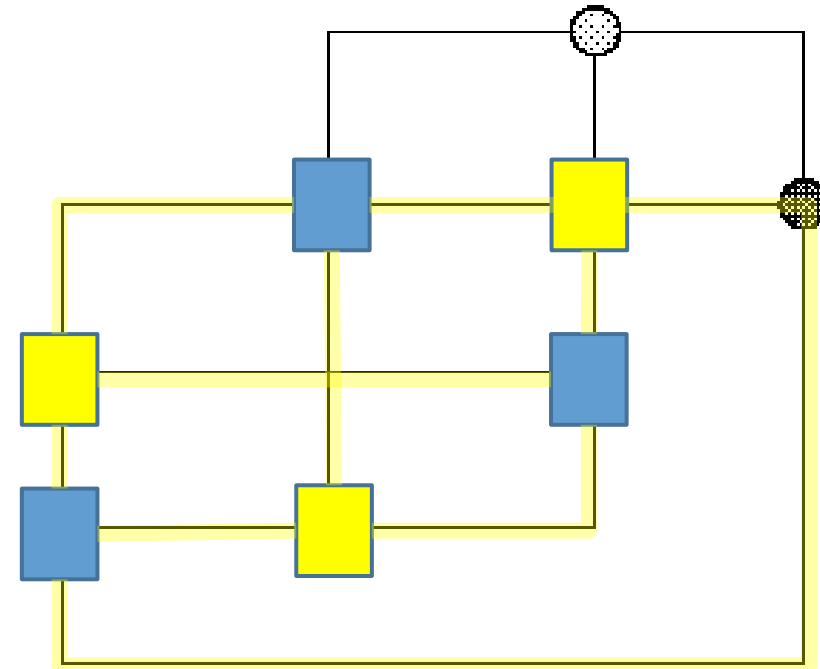
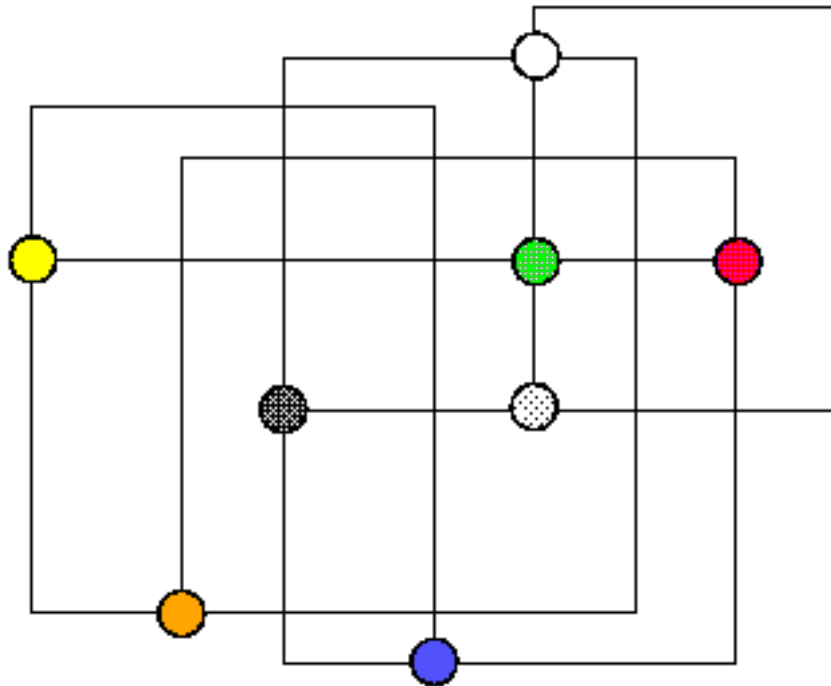
Maximize symmetry

Uncover clusters

Maximize continuity (angle) along paths

Minimize edge crossings

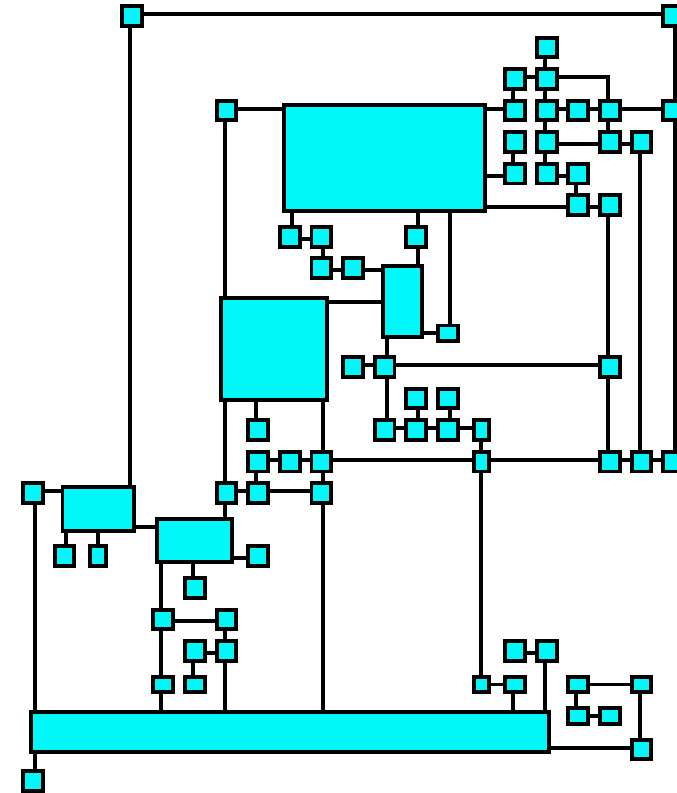
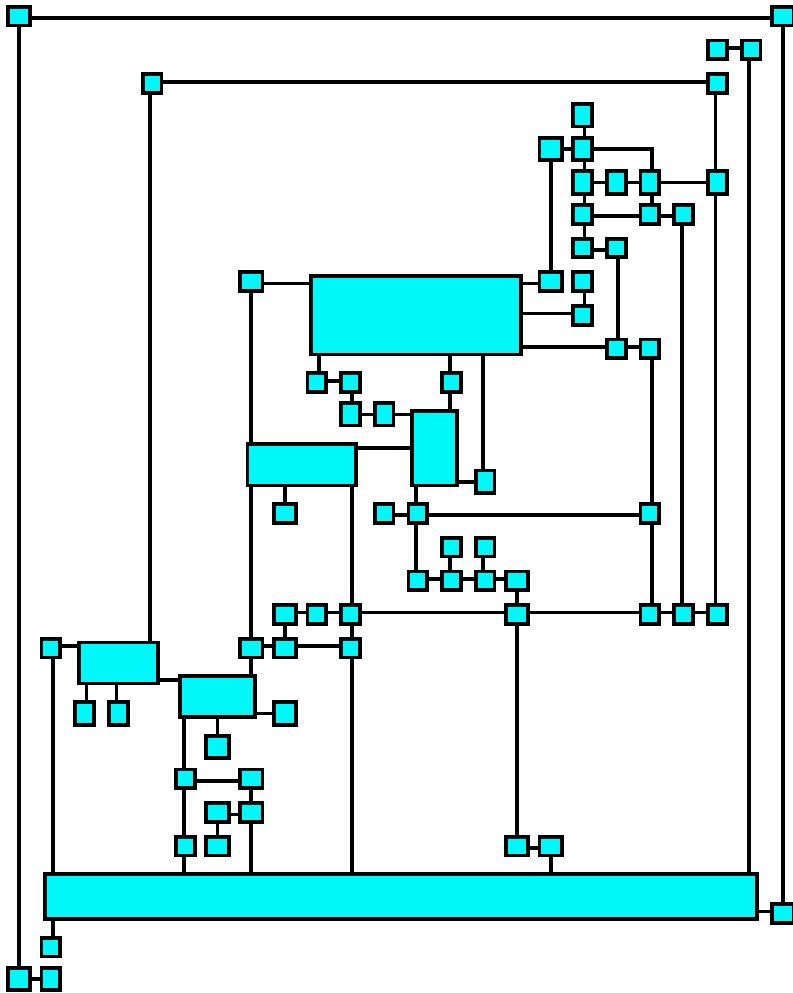
21



In this example the graph is actually **non-planar**, i.e. it cannot be drawn without edge crossings. (Kuratowski's Theorem, $K_{3,3}$)

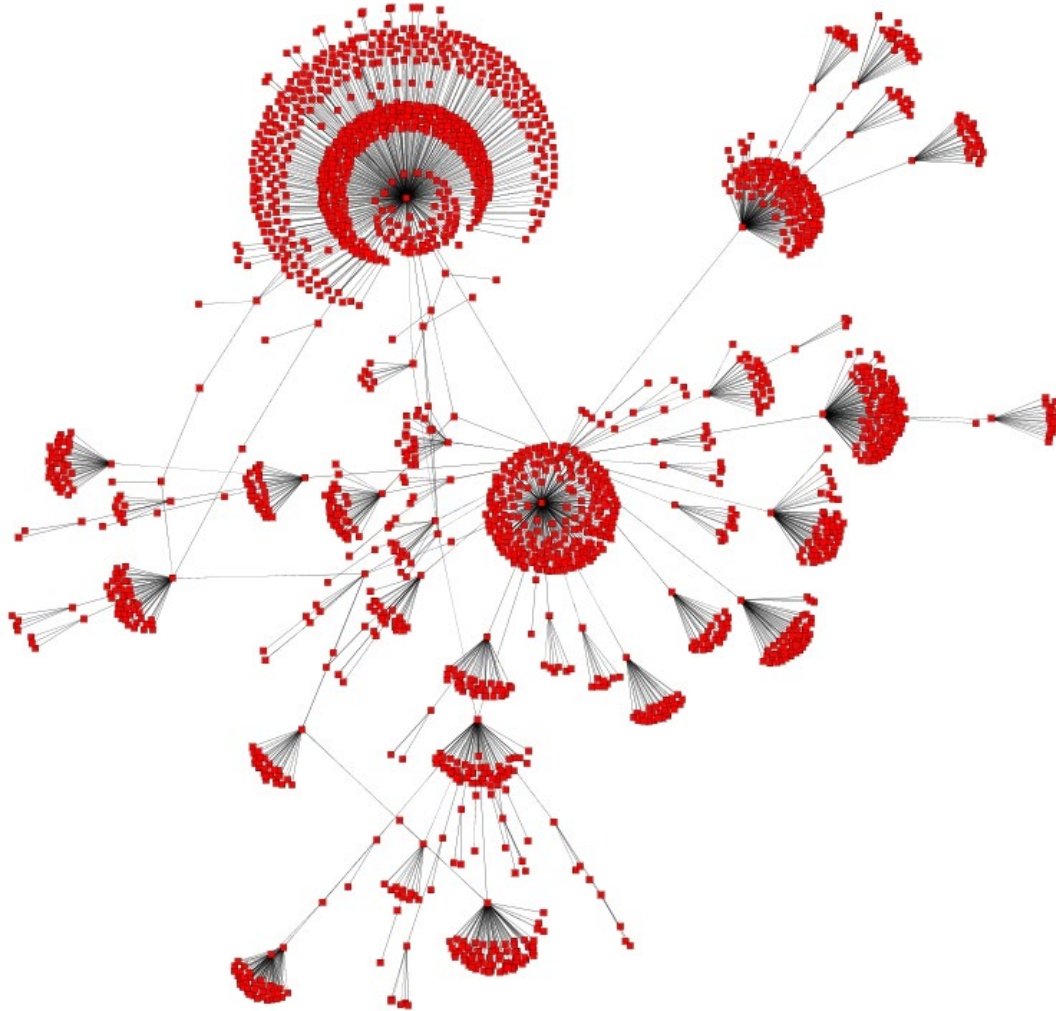
Minimize drawing area/edge length

22



Uncover clusters

23



Cluster = dense subgraph

Clique = complete subgraph

Chapter
5.4

Graph layout algorithms

Category	Percentage
Very bad	10%
Bad	89%

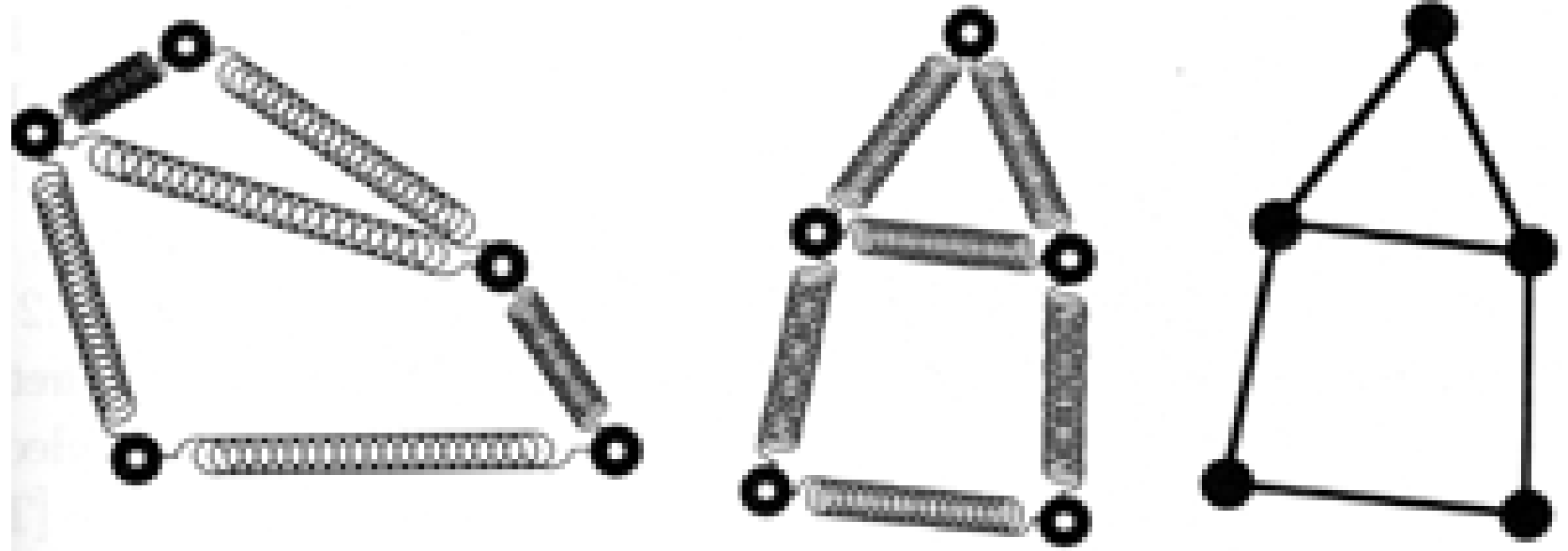


Patrick Braun and Martin Puppe, Structural dependencies in JFtp (Praktikum, Uni Trier)

Force-Directed Layout

26

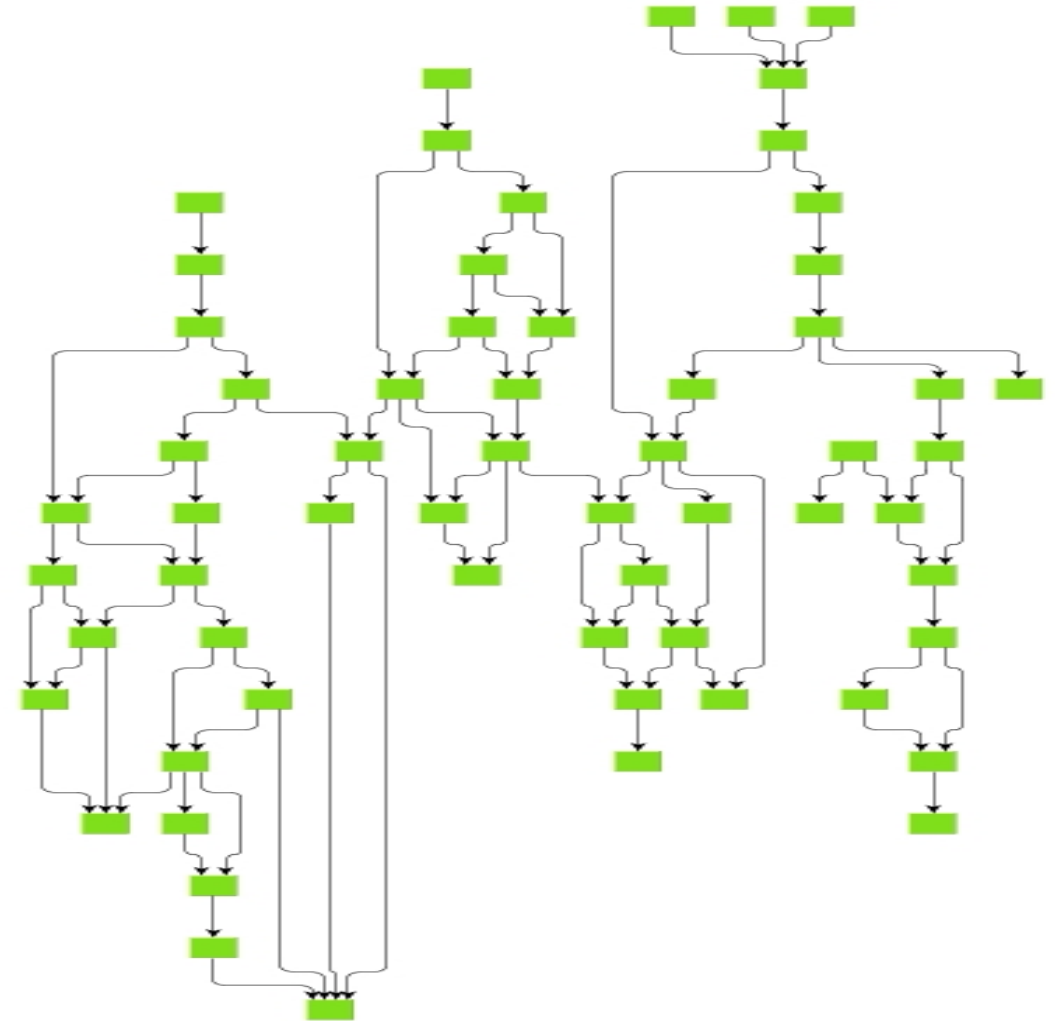
- Springs
- Hooke's law
- Physical simulation
- e.g. Simulated Annealing
- Extensions:
 - Magnetic fields
 - gravity



Hierarchical Layout

27

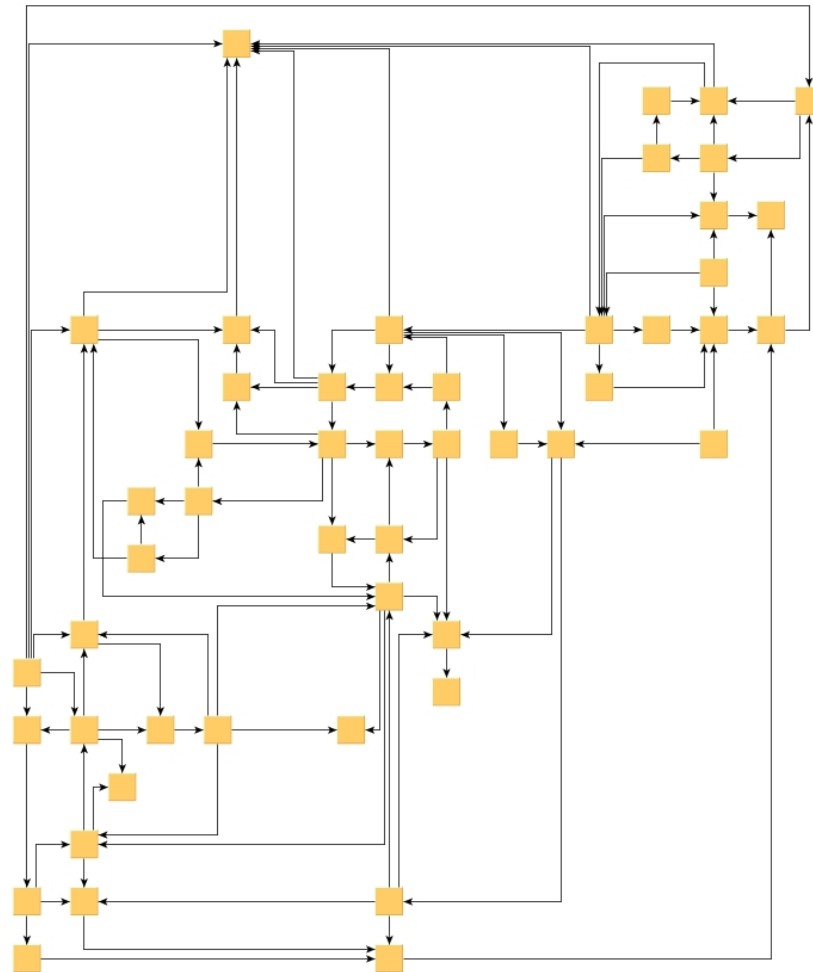
- DAG = Directed, acyclic graphs
- Layers: Topological sorting
- Crossing reduction by reordering layers is NP-complete → heuristics



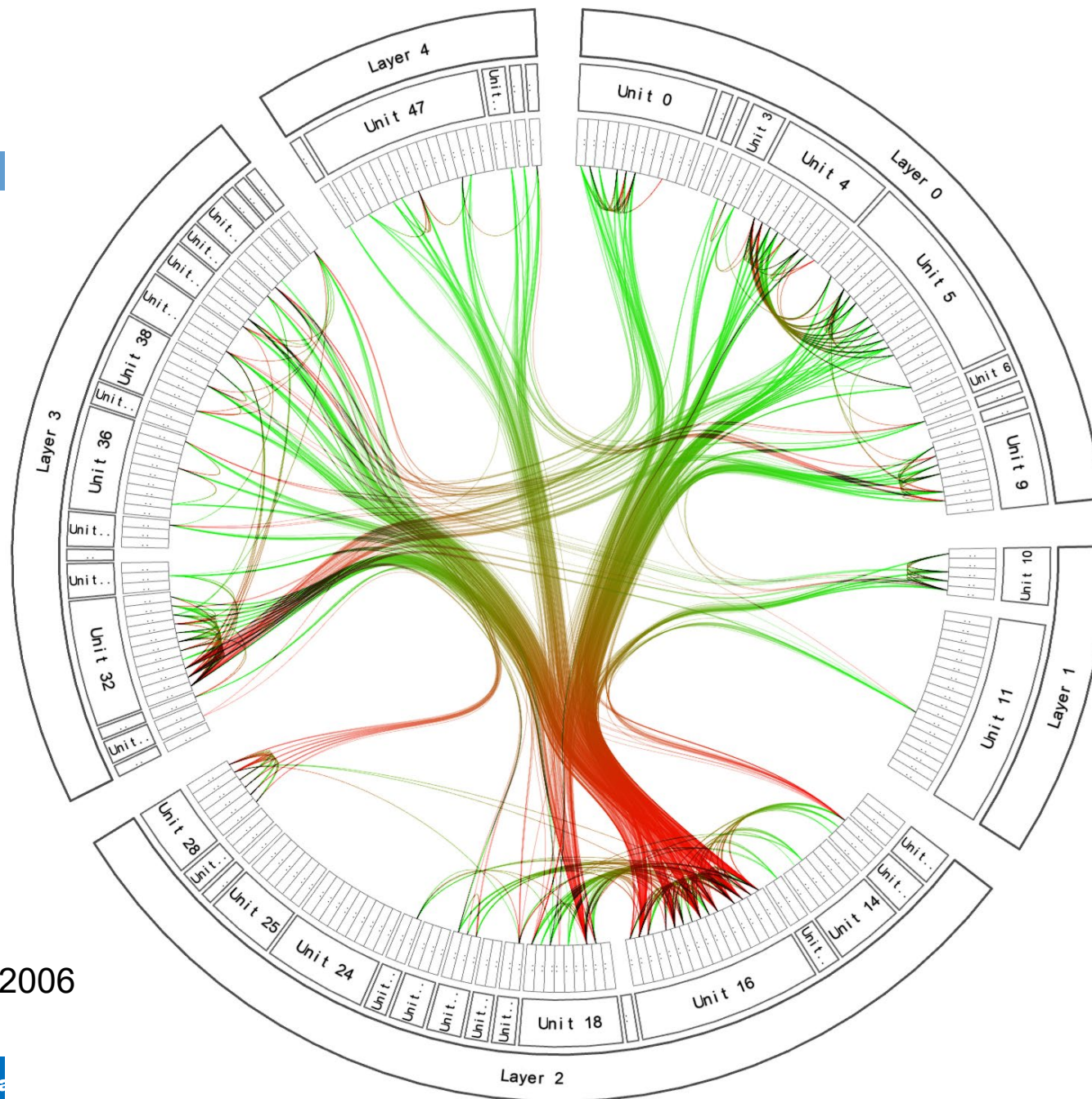
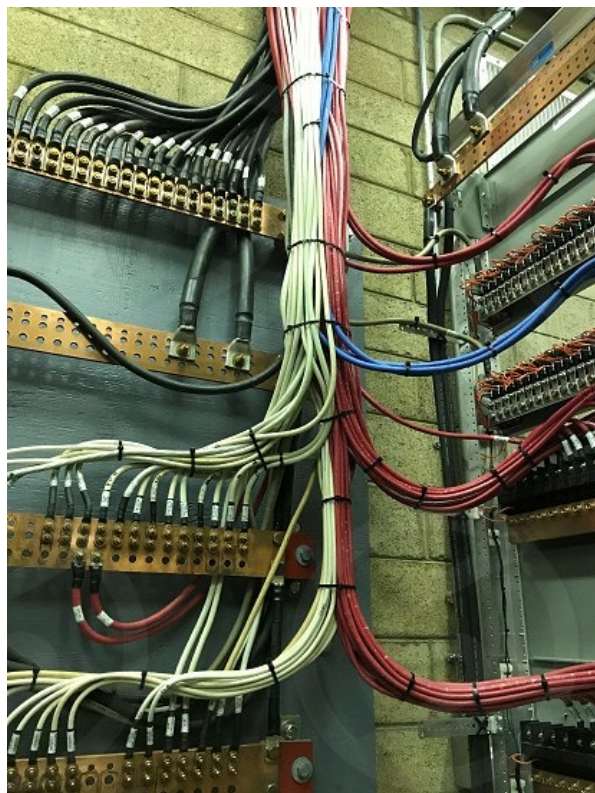


Orthogonal Layout

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Edge Bundling



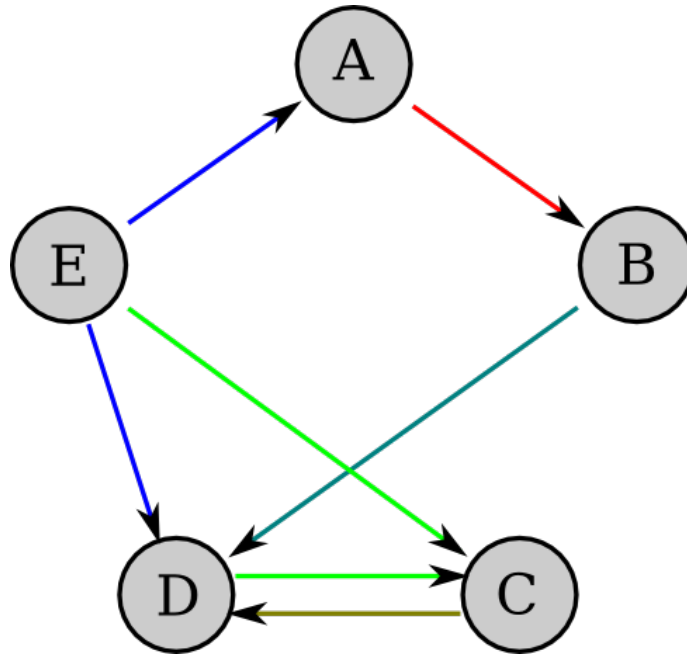
Danny Holten, Hierarchical Edge Bundling, 2006

Chapter
5.5

Matrix visualization of graphs

Representation as adjacency matrix

31



Node-Link

	A	B	C	D	E
A					
B					
C					
D					
E					

Adjacency matrix

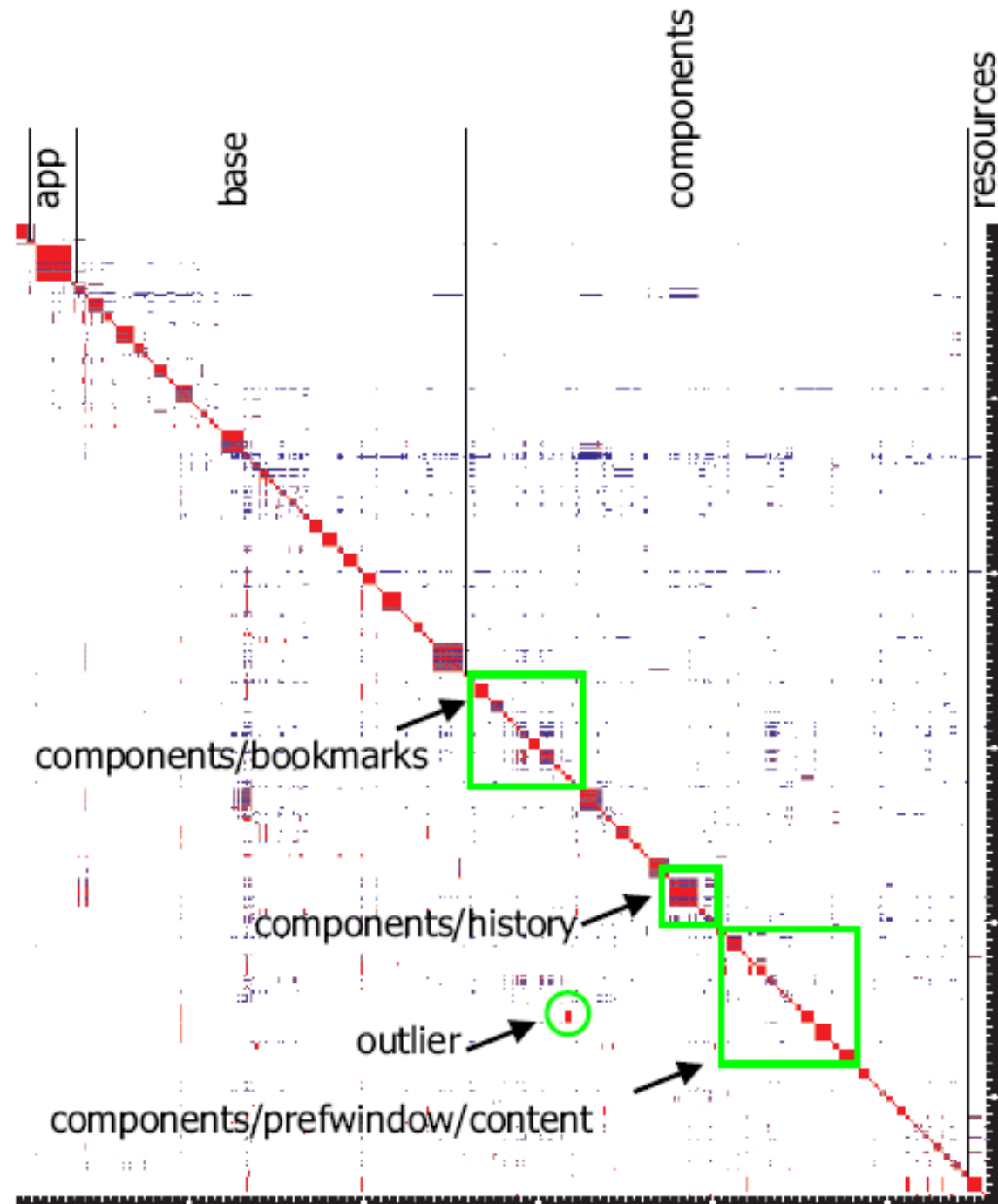
Example

Nodes = Files

Directed edge between file A and B:
When A was changed also B was changed.

Weight of edge indicated by color:
 $\frac{\#(\text{B was changed together with A})}{\#(\text{A was changed})}$

Co-changes of the Mozilla Project



Layout of Matrices

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- Layout problem
 - ▣ linear sorting of columns and rows such that it helps to uncover clusters
 - ▣ NP-complete (good heuristics exist)

	A	D	B	E	C
A					
D					
B					
E					
C					

	A	B	C	D	E
A					
B					
C					
D					
E					

Layout of Matrices

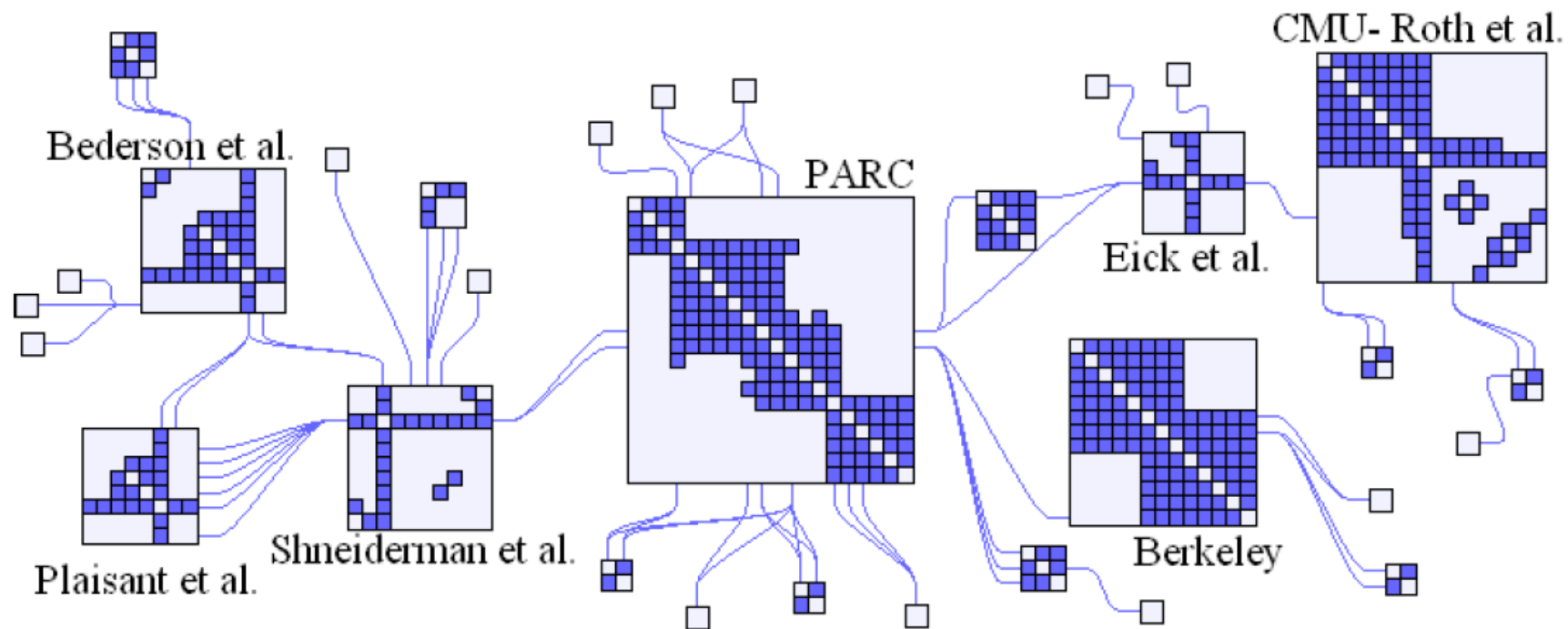
34

- Compound graphs
 - ▣ Hierarchy visualization attached on the sides
 - ▣ Hierarchy enforces a rough sorting
 - ▣ Hierarchy can be used to unfold/fold subgraphs

		a		b	
		A	B	C	
a	A				
	B				
	C				
b	A				
	B				
	C				

Combining node-link and matrix visualizations

35



<http://www.aviz.fr/Research/NodeTrix>

Nathalie Henry et. al, NodeTrix, 2007

Chapter
5.6

Visualization of Dynamic Graphs

Dynamic Graph

37

Temporal sequence of graphs:

$(G_1, G_2, \dots, G_n)$ where $G_i = (V_i, E_i)$

First approach: Animation

38

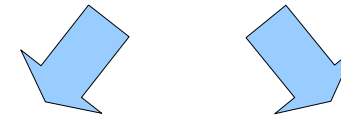
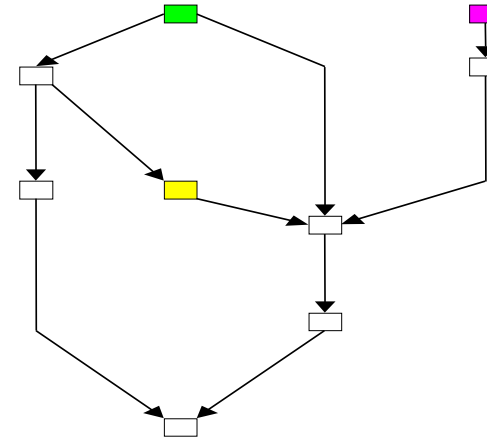
Animation of the sequence of graphs

Local goal: Optimal graph layout

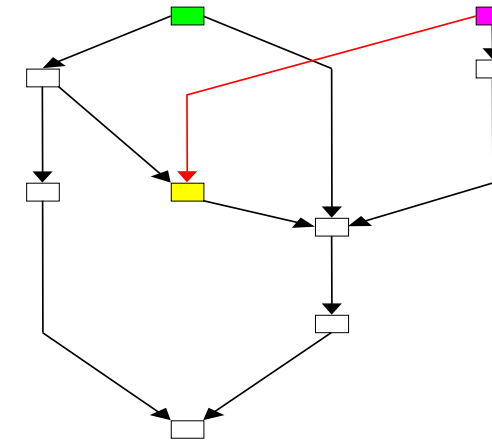
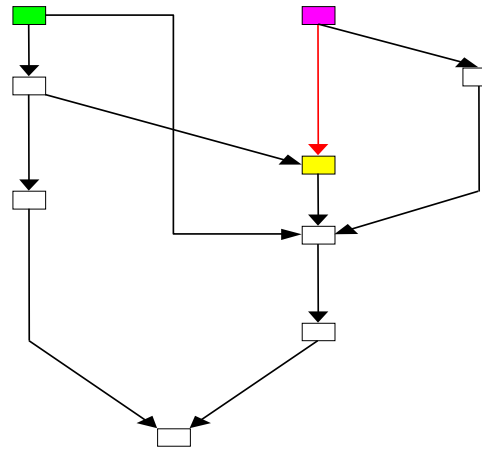
Global goal: Preserving the mental map

(Mental Map = spatial information extracted by looking at a visualization and stored in working memory)

1. Graph



2. Graph



Local optimization of layout

Mental map preservation

Second Approach: Time line

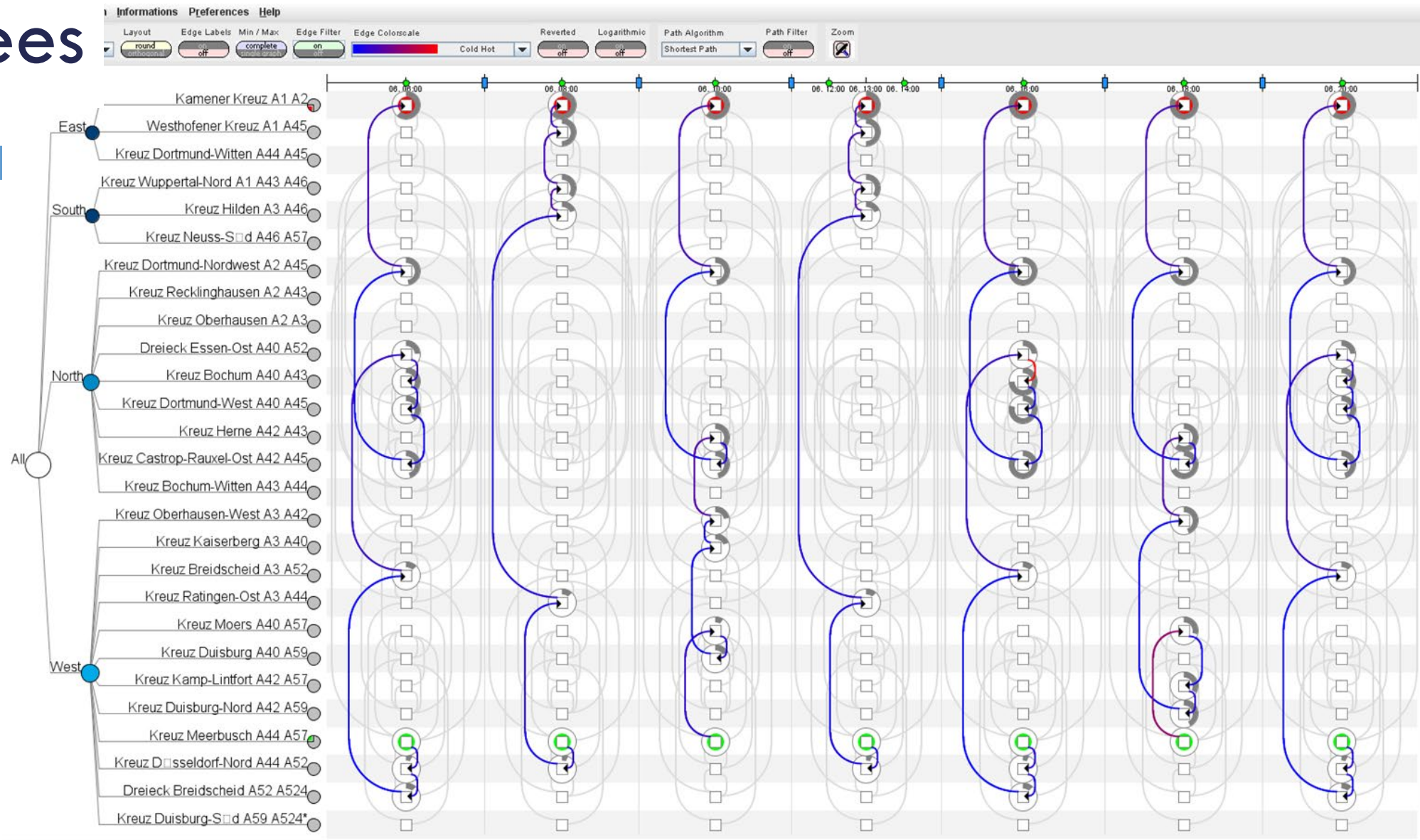
40

Visualization of the sequence of graphs
as a series of static images along a time line.

TimeArcTrees

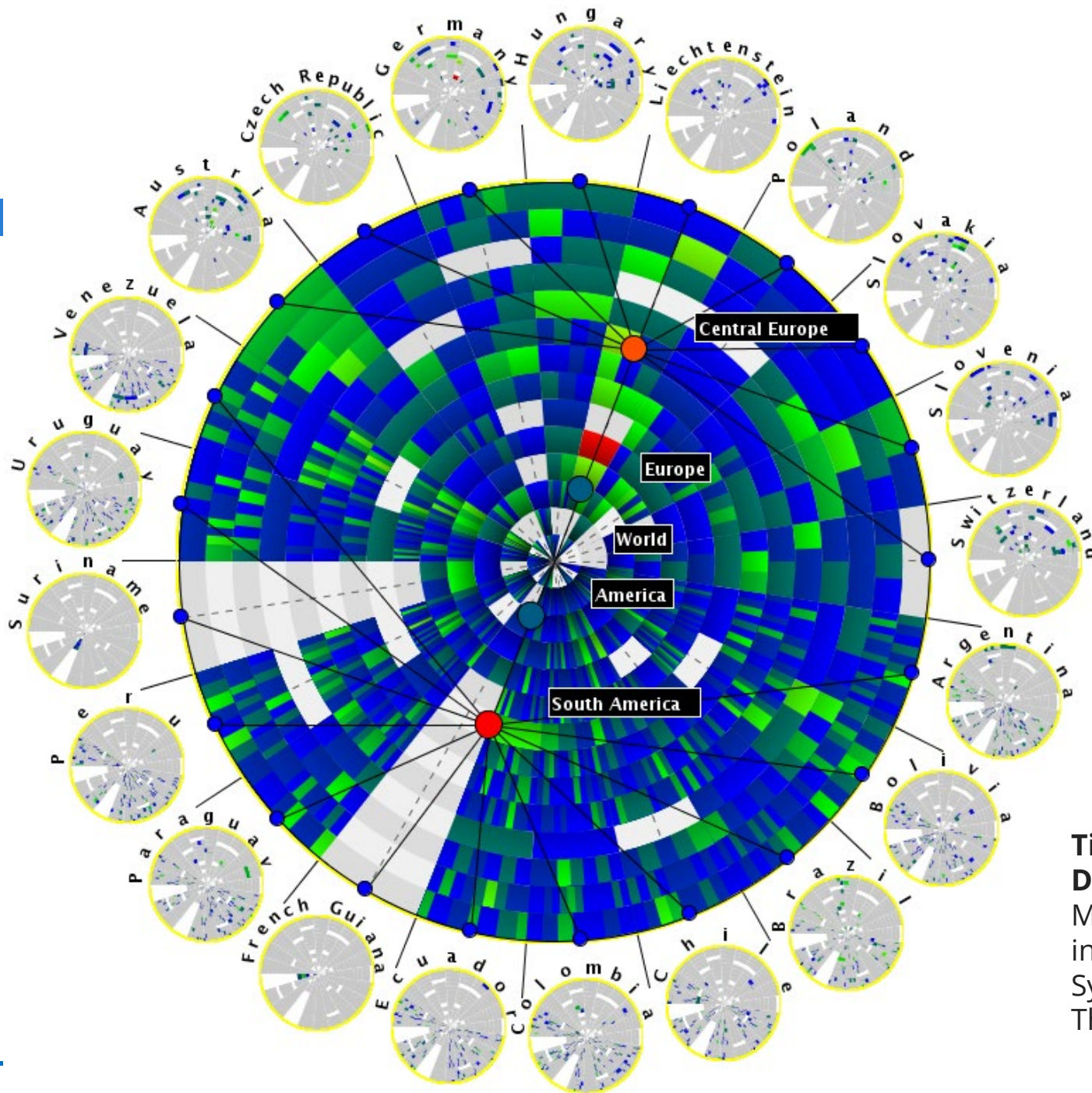
41

Visualizing the Evolution of Compound Digraphs with TimeArcTrees
Martin Greilich, Michael Burch and Stephan Diehl
in Proceedings of 11th Joint Eurographics/IEEE Symposium on Visualization (EuroVis 2009), Berlin, Germany, June 2009.



Temporal change of fastest route from motorway junction Meerbusch (Autobahnkreuz) to motorway junction Kamen (Martin Greilich, Diplomarbeit, Uni Trier)

TimeRadarTrees



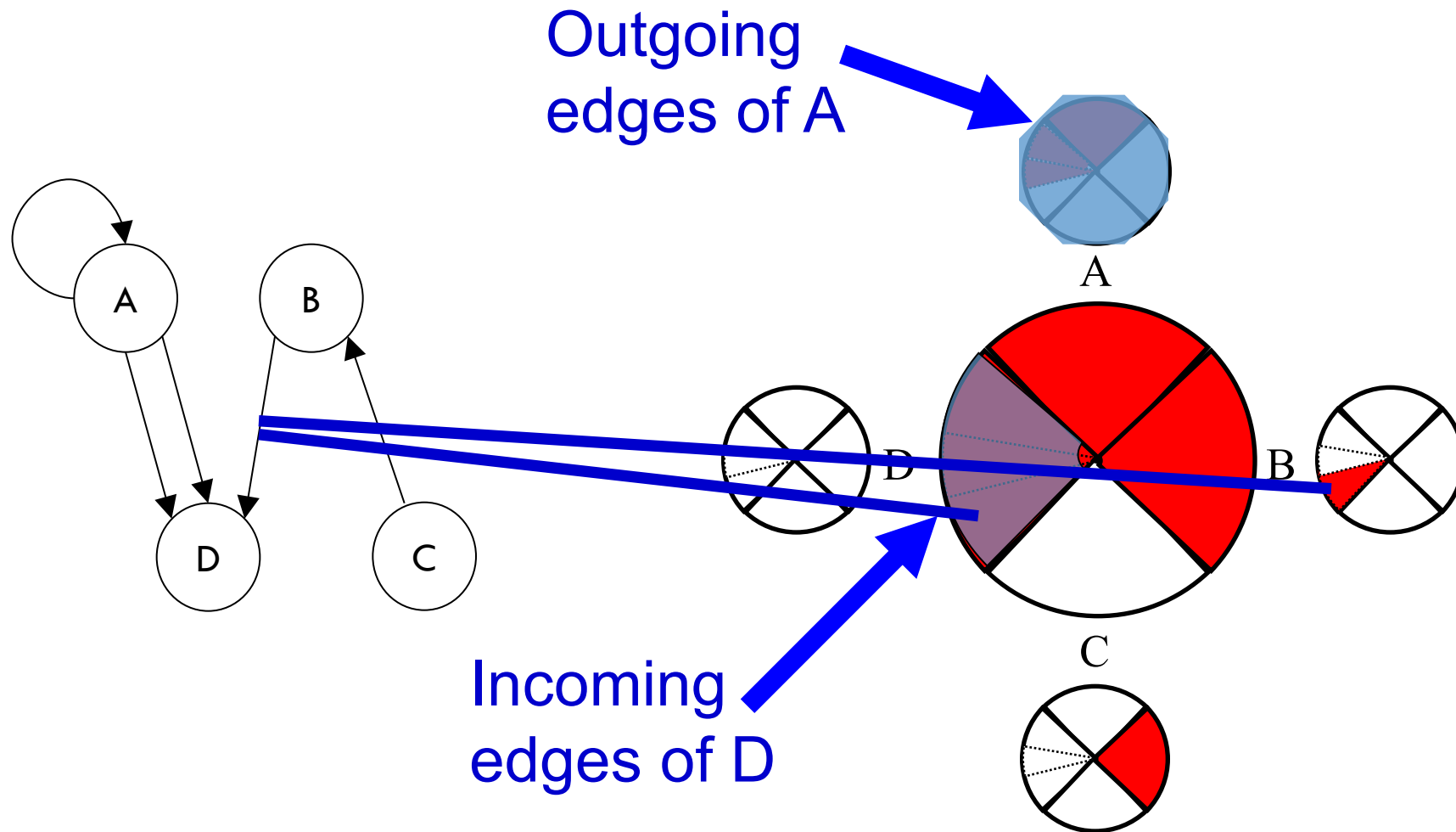
Soccer match graph

- hierarchy: continents, countries
- associations: $A \rightarrow B$
= #goals that team A gave to B
- red: Liechtenstein lost badly in '92 against Germany

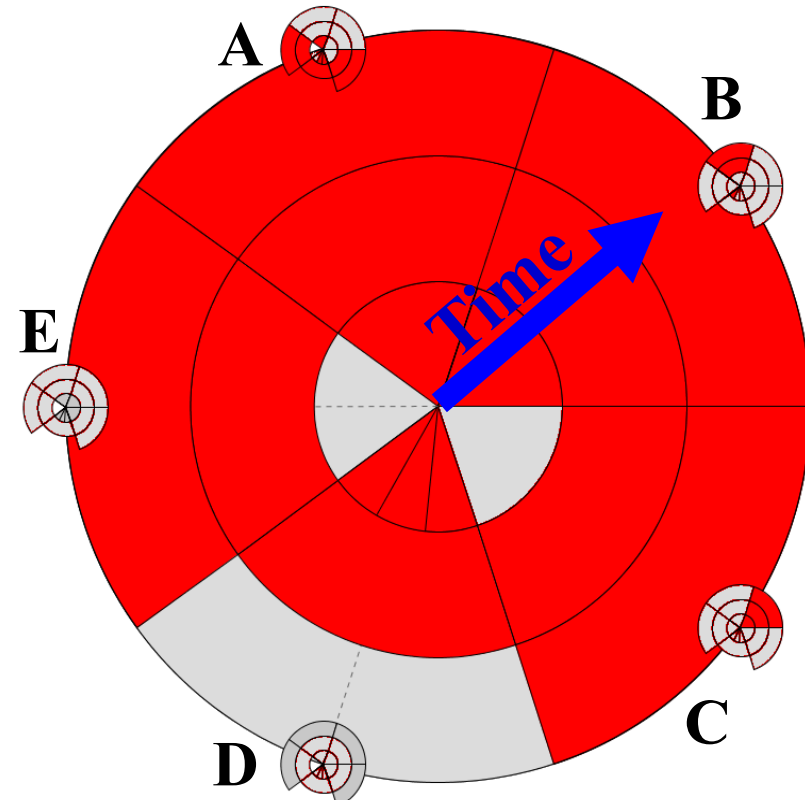
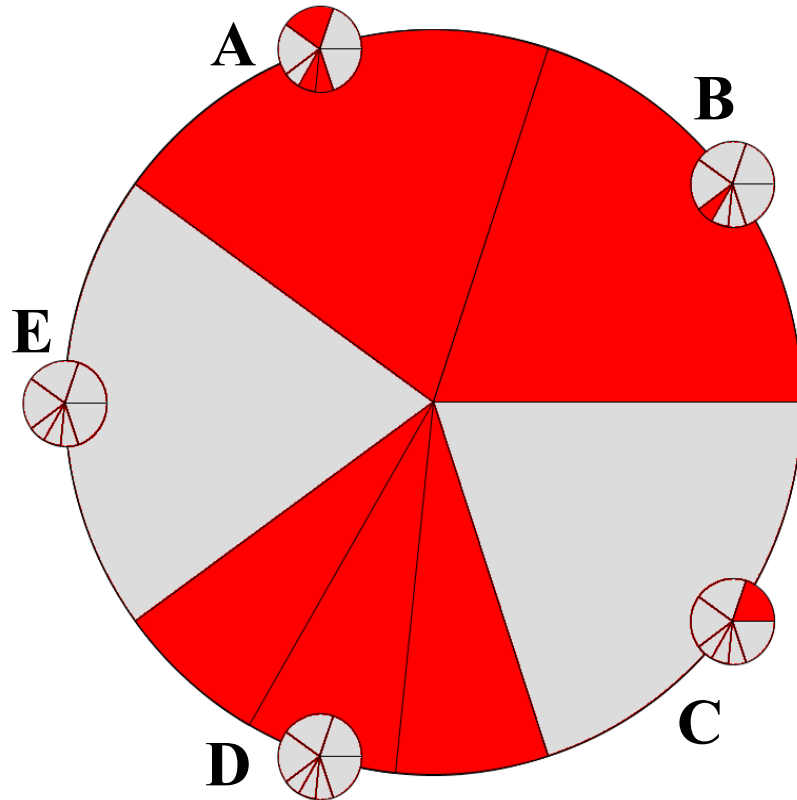
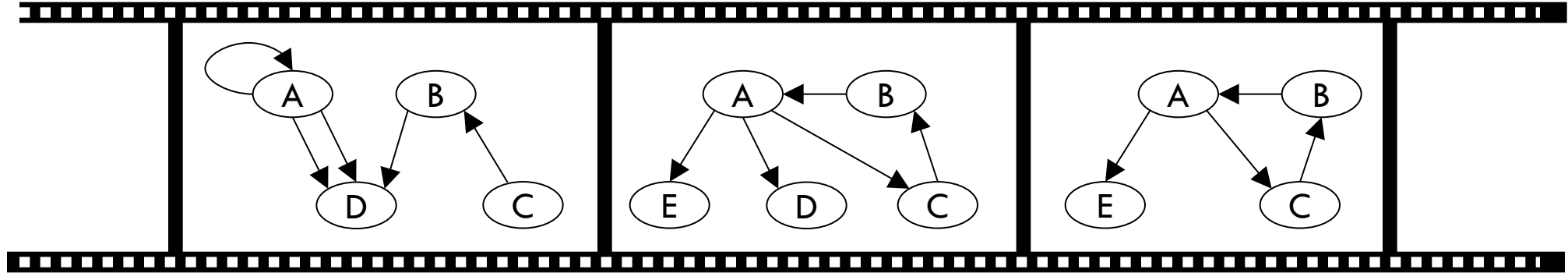
TimeRadarTrees: Visualizing Dynamic Compound Digraphs

Michael Burch and Stephan Diehl
in Proceedings of Tenth Joint Eurographics/IEEE-VGTC
Symposium on Visualization (EuroVis'2008), Eindhoven,
The Netherlands, May 2008.

Edges as Sectors



Sequence of Graphs as Rings

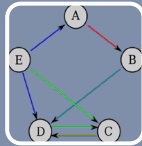


Summary

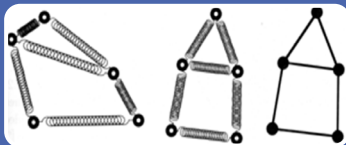
45



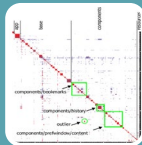
Graph visualization is an important area of information visualization.



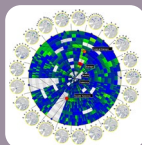
Most common visual representation: Node-link diagram



Layout algorithms optimize aesthetic criteria.



Matrix-based visualizations are an alternative representation, in particular for dense graphs.



Dynamic graphs are mostly visualized using animations or time lines.